Lingua 122 (2012) 494–510

Contents lists available at SciVerse ScienceDirect

Lingua

journal homepage: www.elsevier.com/locate/lingua

The realization of contrastive focus in Peruvian Spanish intonation

Erin O'Rourke *

Department of Modern Languages and Classics, University of Alabama, 219 B.B. Comer Hall, Box 870246, Tuscaloosa, AL 35487-0246, United States

ARTICLE INFO

Article history:
Received 21 August 2009
Received in revised form 9 September 2011
Accepted 15 October 2011
Available online 20 November 2011

Keywords:
Contrastive focus
Intonation
Peak alignment
Peak height
Peruvian Spanish
Language contact

ABSTRACT

In this paper, declarative intonation contours from Peruvian Spanish speakers are compared between pragmatic conditions in order to observe potential differences in the prosodic realization of contrastive focus. To do so, an acoustic feature of intonation, the fundamental frequency (F0), is examined. Since focus in Quechua is realized through the use of evidential markers, it is hypothesized that Cuzco Spanish speakers may behave differently than Lima speakers in relatively less contact with Quechua. Two intonation features have been examined with relation to contrastive focus on the subject: peak alignment and peak height. While Lima speakers demonstrate earlier and higher peaks on the subject, similar to previous reports for Madrid and Peninsular Spanish, some Cuzco speakers use fewer prosodic cues, while others rely on post-focal deaccenting alone. Also, Cuzco speakers appear to behave similarly according to whether or not they are bilingual in Spanish and Quechua. This research demonstrates how focus in realized differently in two regional varieties of Peruvian Spanish and adds to our knowledge of how language contact may contribute to the development of alternate intonation patterns in Spanish.
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1. Introduction

This paper addresses how contrastive focus is expressed in Peruvian Spanish intonation. In particular, the fundamental frequency (F0) will be examined as an acoustic measure of the intonation contour. An element under focus is considered to be emphasized or highlighted. If all information is highlighted equally, the utterance is considered to be under 'broad focus'; if a portion is emphasized, the element is under 'narrow focus'; and if a portion is highlighted in comparison to another element in the utterance or discourse, the element is under '(narrow) contrastive focus' (Ladd, 1980, §4.2, among others). Previous research on focus in Spanish as well as in other languages has demonstrated that certain prosodic features may be realized differently in utterances with narrow or contrastive focus compared to more neutral, broad focus utterances.[1] For example, increases in duration may be observed at the syllable, word, or sentence level (Face, 2002; Ito, 2002; Kim and Avelino, 2004; de la Mota Gorriz, 1997; Shih, 1997, 2000; Sityaev and House, 2003; Smiljanić, 2002; Smiljanić and Hualde, 2000).[2] For Peninsular Spanish, the tonal peak associated with the stressed syllable under narrow and/or contrastive focus in

* Corresponding author. Tel.: +1 205 348 6046; fax: +1 205 348 2042.
E-mail address: eorourke@bama.ua.edu

[1] For this description, broad focus is understood to be present when the prior discourse does not restrict the expected domain of the new material. That is, all information is considered to be novel. Narrow focus, on the other hand, obtains as a response to a question (direct or implied) that requires new information for a portion of the utterance not previously available in the discourse context. Finally, contrastive focus is understood to be a subtype of narrow focus which counters the previous proposition, and provides information that is considered to be a correction to the previous discourse context.

[2] See Ito (2002) for further description of durational effects under focus in several languages.

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doi:10.1016/j.lingua.2011.10.002

non-final position appears to align earlier than in broad focus (Face, 2001; de la Mota Gorriz, 1997).[3] In European Portuguese, the peak is realized later in contrastive focus: that is, the peak is aligned within the tonic syllable as opposed to before the tonic syllable in broad focus (Frota, 1998). Japanese also shows later peak alignment, in this case occurring later on the post-tonic syllable (Ito, 2002). In other cases, such as in Dominican Spanish, the low tone is shown to align earlier in the stressed syllable under contrastive focus (Willis, 2003a, 2003b).

Several studies have also observed changes in the height of peaks in utterances with narrow and contrastive focus. Specifically, words with focus may show higher tonal peaks, post-focal words may be markedly lower in tonal height, while pre-focal peaks may be similar to peaks in broad focus utterances[4] (Barjam, 2004; Cabrera Abreu and García Lecumberri, 2003; D'Imperio, 2001; Domínguez, 2004; Face, 2001; García Lecumberri, 1995; Ito, 2002; de la Mota Gorriz, 1997; Shih, 1997, 2000; Smiljanić, 2002; Xu, 1999). Height effects were sometimes reported for all speakers, whereas in other cases, the behavior was observed for some speakers but not for others. Frota (1998) found variation in European Portuguese between and within speakers, and notes that "the raising of the H (and consequently, the larger fall) is an optional gradient feature that the focus nucleus may show, probably linked to a gradience of emphasis" (1998:218). In other cases, an effect of peak height was not observed in focus conditions (Sityaev and House, 2003).

The remainder of the paper is structured as follows. Below we discuss research on intonation and focus, with specific attention to language contact and research on Spanish. In section 2 we discuss focus in Quechua and Spanish. Then, in section 3 we describe the present experiment, including participants, the data set used, how measurements were taken, and the hypotheses to be tested. Section 4 provides a presentation of the results, including a comparison of the focus strategies employed across groups of speakers.

1.1. Research on intonation and focus

Cross-linguistic and cross-dialectal studies have demonstrated differences in the recruitment of focal strategies for the same utterance type. In addition, there may be multiple cues to indicate a given pragmatic contrast, and different combinations of these cues. In this way, one or more intonation cues may be found to express focus, similar to other uses of multiple cues in language, for example, the realization of stress via (increased) syllable duration and/or intensity, differences in vowel quality, etc., along with changes in fundamental frequency. Gussenhoven (2002:280) summarizes some of the recent research on the variety of strategies observed in different languages to express focus, including alignment specifications, use of a special pitch accent, as well as deaccenting, in which either no or a decreased tonal movement is observed related to the stressed syllable (see Ladd, 1980, §3; Bolinger, 1986, §7 for discussion of deaccenting). It should be noted that some of these features may be employed differently, even under broad focus conditions across language varieties. For example, alignment differences are often found. In declaratives in Donnegal Irish, late peaks are observed (L* + H) compared to Connaught Irish, which shows early peaks (H*) (Dalton and Chasaide, 2003). Similarly, two varieties of Scottish show alignment distinctions: post-tonic prenuclear[5] peaks are recognized as characteristic of Orkney speech, while tonic prenuclear peaks are perceived as typical of Shetland speech (van Heuven and van Leyden, 2003; van Leyden, 2004). Atterer and Ladd (2004) show more fine-grained, yet still significant differences in the realization of peaks between varieties of German, with valleys and peaks in Southern German being realized later than valleys and peaks in Northern German. Analyzing Serbian and Croatian, Smiljanić and Hualde (2000) and Smiljanić (2002) show that lexical alignment distinctions present in Belgrade may still shift under contrastive focus. Also, Zagreb speakers, who no longer exhibit lexical alignment contrasts, likewise demonstrate earlier peaks with contrastive focus, even for one speaker who already shows tonic alignment of peaks under broad focus (2002:77).

In some instances, the reason for and consequence of alignment differences is discussed in terms of possible language contact with other languages of the region, for example Irish in contact with different varieties of English, and Scottish in contact with Scandinavian, Celtic languages and historically, Norn (van Heuven and van Leyden, 2003). Research in another contact situation, that of Basque in northern Spain, also shows the maintenance of the characteristic Basque tonic alignment of prenuclear peaks in Basque broad focus utterances in a region where post-tonic peaks are found in Spanish. These bilingual speakers of Basque and Spanish show a late rise in Basque only to signal a topic in utterances with focus on subsequent words. In this sense, more than one intonation system can be found in the same geographical area (Ito et al., 2003). In this way, studies of language contact can show how contours specific to one particular group can develop over time.

[3] The extent to which these alignment distinctions are categorical needs to be further understood. In a study examining tonal alignment in phrase-final position in English (Pierrehumbert and Steele, 1989), listeners repeated stimuli that were resynthesized along a peak-placement continuum. However, their imitations of these stimuli were produced in a bimodal distribution, suggesting that at the abstract level there is a categorical distinction of peak alignment. The number, size, and locations of categories (e.g., pre- vs. post-tonic), may differ across languages, which in turn may affect how these categories interact and are interpreted in language contact situations.

[4] These differences in height may produce relative differences in scaling in the given utterance, which may function along with or separately from differences in overall pitch range employed under focus.

[5] The term 'prenuclear' is used to refer to non-final peaks or pitch accents within an utterance that precede the 'nuclear' pitch accent, which often occurs in utterance-final position (e.g., in English). The term 'tonic' is used here to refer to the stressed syllable, so that peaks occurring after the end of the stressed syllable are 'post-tonic'. In this sense, then tonal peaks occurring after the end of the stressed syllable (i.e., post-tonic) may be prenuclear if they occur on non-final words in an utterance.

Given the necessarily conditions for wider distribution to non-contact varieties, the intonation of languages in contact may contribute to long-term language change (see Thomason and Kaufman, 1988 for discussion).

Turning to Spanish, there are studies that show alignment of prenuclear peaks with the stressed syllable in broad focus utterances (e.g., Elordieta, 2003 for Lekeitio Spanish, and Colantoni and Gurlekian, 2004 for Buenos Aires Spanish). Elordieta (2003) notes that these Basque and Spanish bilinguals are not producing contours previously reported for Madrid and Peninsular Spanish nor do they employ the H* + L prenuclear pitch accent attributed to Basque. Rather, these bilinguals are employing unique contours not found in either source language of the region, that is L + H*. In Buenos Aires Spanish, prenuclear peaks are observed to show tonic alignment under broad focus, a realization that is attributed to a reanalysis of an Italian focal pitch accent that was applied during a prolonged period of language contact in Argentina with Italian and the Italian-influenced Spanish variety, Lunfardo (Colantoni and Gurlekian, 2004; see also Colantoni, 2011). McMahon (2004) notes that cases such as those found in Buenos Aires discussed by Colantoni and Gurlekian demonstrate the importance of tracking changes in intonation in situations of language contact as well as in non-contact situations.

Spanish as spoken in Peru presents itself as another potential situation in which differences may be found in the realization of focus through intonation. Quechua, one of the most widely spoken indigenous languages in Peru, marks focus morphologically, unlike Spanish, which does not possess a specific suffix for focus. And, since Quechua is spoken to varying degrees according to geographical region, comparisons can be made between monolingual Spanish speakers from Lima, and Cuzco speakers with different levels of knowledge and use of Quechua. Therefore, it is important is to determine how focus is realized in Peruvian Spanish compared to other Spanish varieties, and the extent to which knowledge of Quechua appears to influence the number and type of intonation strategies employed to express focus. In this way, the current research project adds to what is currently known about intonation in one variety of Latin American Spanish—Peruvian Spanish, and contributes in a broader sense to prosodic research on Spanish in contact with indigenous languages, and languages in contact in general. In addition, this project demonstrates individual variation in the expression of focus while also observing reoccurring patterns within groups.

2. Focus in Quechua and Spanish

In the introduction above, prosodic manifestations of focus in Spanish and several other languages were described. In this section, we compare the linguistic resources Quechua and Spanish employ to express focus. Then the findings of intonation research for Spanish are reviewed in more detail according to the features to be examined in this current study: peak alignment and peak height.

In Quechua, focus is marked by adding a suffix to the element under focus. This suffix may be one of a set of 'evidential' or 'validator' suffixes that underline the truth-value of the utterance, as shown in (1) and (2).[6]

(1) *Pidru-n* *wasi-ta* *ruwa-n*
Pedro-VAL house-ACC make-3SG
'It is Pedro that builds a house'.

(2) *Pidru* *wasi-ta-n* *ruwa-n*
Pedro house-ACC-VAL make-3SG
'It is a house that Pedro builds'. (Muysken, 1995, underlining added)

In the existing descriptions of Quechua, there is little or no information given regarding the type of prosodic features that may accompany the expression of focus. An exception is the description by Cole (1982) of Quechua as spoken in Imbabura, Ecuador, which characterizes the non-final peak under contrastive focus as being higher than peaks in the same position in broad focus (1982:210-211).

In O'Rourke (2005), a small set of utterances with the evidential suffixes *–mi/m* were analyzed. The preliminary observation was that the peaks associated with elements bearing this morphological marking did not appear to be realized higher than elements that did not. Given the general absence of more detailed descriptions in the literature, including pedagogical texts, it appears that the use of intonation cues for focus, if present in Quechua, may be considered an additional strategy that may appear alongside the strategy found at the morphological level. The way in which intonation may play a role in marking focus in Quechua is left open for future research.

Unlike Quechua, focus in Spanish is not accompanied by a specific morphological marker. However, there are some constructions in Spanish which may be considered to mark focus, such as the use of focalizing *ser* 'to be', for example, in Colombian Spanish, among other varieties (Méndez Vallejo, 2008, and references cited therein). Domínguez (2004) examines the use in Spanish of syntactic constructions and prosodic cues to indicate focus on an element, and the interface

[6] As Muysken (1995:380–381) points out, an alternate interpretation of (1) is 'Pedro builds a house' (without focus) since the initial position is taken as a default position for evidentials compared to (2), which shows the evidential on the second element and is therefore interpreted as having focus.

Let them give grammaticality judgements for diff focus constructions

between the two. A distinction is made between contrastive focus realized in the left periphery, contrastive focus *in situ*, and information focus (i.e., narrow non-contrastive focus) found in the right periphery. Of special interest for the current study is contrastive focus *in situ* in which the subject remains in initial position, which Domínguez notes is prosodically licensed.[7]

In this study of Peruvian Spanish, the intonation contour in utterances with contrastive focus on the initial subject peak are examined according to peak alignment and peak height (both absolute and relative). Each of the features of interest for the current study is reviewed in turn below with respect to Spanish. Other features that are not examined, but may possibly still be relevant, are: the duration of all segments (stressed and unstressed syllables, word durations, sentence duration); tonal range (both the global range of the utterance, and local tonal range of the pitch accent excursion); the initial and final F0 height; the use of an intermediate phrase boundary tone (e.g., a post-focal low target (L-), or a high tone (H-) followed by a pause); as well as changes in intensity and vowel quality in the two pragmatic conditions.[8]

First, previous studies on tonal alignment in Spanish show that prenuclear peaks are described as aligning within the stressed syllable under narrow and contrastive focus for Madrid and Peninsular Spanish (Face, 2001; de la Mota Gorriz, 1997, among others). The underlying phonological representation of these tonal targets lies beyond the scope of the present discussion, although an examination of research on Spanish and implications appears in Hualde (2002) and Face (2002).[9] However, there are examples of varieties that already show this type of stressed-syllable alignment under broad focus: Lekeitio Spanish (Elordieta, 2003), Buenos Aires Spanish (Barjam, 2004; Colantoni and Gurlekian, 2004), and Mexico City Spanish (Kim and Avelino, 2004). Previously, alignment of peaks within the tonic syllable was also found in broad focus declaratives for some Cuzco Spanish speakers (O'Rourke, 2004, 2005). Barjam (2004) found that even though broad focus peaks in Buenos Aires Spanish were already aligned within the stressed syllable, peaks with contrastive focus were still aligned significantly earlier. Since variation was observed according to speaker, this result is taken as an indication that focus is marked by relative rather than absolute timing differences (2004:71). On the other hand, in Dominican Spanish, it was low tone alignment in contrastive focus utterances that was shown to be earlier, whereas the peak remained on the post-tonic in this utterance type (Willis, 2003a, 2003b). Willis notes that this tonal configuration for contrastive focus in Dominican Spanish, 'an early low and late high', is in fact similar to the configuration observed for Madrid and Mexican Spanish broad focus declaratives. In this way, data from Dominican Spanish demonstrates a cross-dialectal difference in the recruitment of tonal alignment to indicate a different pragmatic meaning. For Peruvian Spanish, another cross-dialectal difference may be anticipated. If Lima and some Cuzco speakers show tonic alignment of prenuclear peaks only in focus conditions, then these speakers may perceive the same type of pragmatic meaning (i.e., contrastive focus) when listening to those Cuzco speakers who show tonic prenuclear alignment even under broad focus conditions. While the present study does not address listener perceptions, this research will begin to establish what types of behavior are produced by both Lima and Cuzco speakers under different focus and non-focus conditions.

Second, the relationship of peak heights has been examined in different pragmatic conditions. Face (2002) summarizes previous findings on differences in peak height: Toledo (1989) showed variation in the use of significantly higher peaks in narrow and contrastive focus across and within speakers for Buenos Aires Spanish; de la Mota Gorriz (1995, 1997), García Lecumberri (1995) showed significantly higher peaks under contrastive focus, although the latter showed variation across speakers in Peninsular Spanish; and Face (2000) showed no significant differences in peak heights between narrow and broad focus for Madrid Spanish. An examination of contrastive focus in Madrid Spanish, however, shows significantly higher peaks for those utterances that employed pitch accents with post-tonic peaks (L* + H) (Face, 2002).

In sum, we have seen in the introduction that languages may employ one or more prosodic cues to signal focus. In addition, within Spanish the feature of tonal peak alignment under focus was observed to differ in contact varieties, such as Basque-influenced and Italian-influenced Spanish; the feature of tonal peak height under focus has also been found to be employed in some cases but not others for Spanish. In section 2, we noted that in Peru, one of the contact languages of the region, Quechua, marks focus morphologically through the use of evidential suffixes, whereas Spanish typically does not. Therefore, the study described below has been conducted to examine the extent to which Spanish speakers with knowledge of Quechua may demonstrate distinct intonation patterns compared to non-contact varieties. If bilingual speakers are less accustomed to marking intonation prosodically due influence from Quechua, then they may demonstrate either no use of intonation for elements under focus, non-target-like use of intonation, or target-like use of intonation at or below the expected levels of frequency. In the section below, we describe the present study, including participants, data set, and measurements in sections 3.1–3.3, along with predictions of potential outcomes in section 3.4.

ex. of in situ focus

[7] To the question, 'Was it Antonio that saw the accident?' the response in (a) shows the subject in final position, whereas in (b) it appears in initial position. Both are considered appropriate responses (Domínguez, 2004:14–15).

(a)	No, lo vio [$_F$ Carlos]	(b)	No, [$_F$ Carlos] vio el accidente
	No it saw Carlos		No Carlos saw the accident
	'No, Carlos saw it'.		'No, Carlos saw the accident'.

[8] See Nibert (2000) and Face (2001, 2002) for an examination of the use of intermediate phrase tones in utterances with contrastive focus in Spanish.

[9] Much subsequent work has been conducted in the area of Spanish intonation and its relevance within Romance (e.g., research in Colantoni and Steele, 2008; Elordieta and Vigário, 2007; Face, 2004; Prieto et al., 2007, among many others).

3. Experiment

3.1. Participants

The participants in this study were Peruvian Spanish speakers from either Lima or Cuzco. The speakers ranged in age from 18 to 39 years, and had received or were enrolled in a post-secondary degree program. In order to control for any differences in gender, only male speakers are included in this data set; recordings from female speakers are needed in order to further generalize the findings reported here.

A total of 21 speakers have been analyzed for this study, including six Lima speakers whose parents were also from the Department of Lima, and seventeen Cuzco speakers whose parents were likewise from the Department of Cuzco. The Lima speakers have been assigned the identification code L01–L06. The Cuzco speakers can be further divided into subgroups of nine Spanish monolinguals (C01–C09), five Quechua-Spanish bilinguals (C21–C25), and three native Quechua speakers (C31–C33). While ideally there would be even numbers of speakers in all four groups (e.g., five in each group), additional speakers have been included to examine in greater detail the heterogeneity of the Spanish monolinguals. Also, adding more native Quechua speakers to this data set would also be useful in order to better characterize the behavior of this group.

3.2. Data set

Each speaker read a list of utterances that were preceded by a prompting question designed to elicit contrastive focus on the subject. For example, the question *¿La tienda mandará los violines?* 'Will the store send the violins?' precedes the target answer, *No. Su familia mandará los violines*. 'No. His family will send the violins'. In this case, *familia* 'family' is taken as contrastive to *tienda* 'store', both in subject position. Only those utterances with contrast placed on the subject in initial position are examined for this current study (as opposed to contrast placed in other positions, such as on the object, e.g., *guitarras* 'guitars' vs. *violins* 'violins'). The results from contrastive focus on the subject are compared with results from question-and-answer pairs designed to elicit broad focus, using the prompting questions *¿Qué pasa/pasaba/pasará?* 'What is happening/was happening/will happen?', as previously reported in O'Rourke (2004, 2005). In the example above, the prompting question *¿Qué pasará?* 'What will happen?' was used to elicit the broad focus response *Su familia mandará los violines* 'His/her family will send the violins'.

There were 24 utterances with contrastive focus on the subject that were recorded per speaker (12 sentences x 2 productions). Five Lima speakers and 17 Cuzco speakers performed this task. For the sixth Lima speaker, who performed a similar task, there were 30 utterances with subject contrastive focus (15 utterances x 2 productions). A total of 558 utterances were examined for this data set. The contrastive focus data are compared to an equal number of utterances produced in response to broad focus questions. In these sentences, different syllable types (open and closed) and stress patterns (antepenult, penult, and final) were present. For the analysis of tonal peak alignment, only tokens with open syllables and stress on the penultimate syllable were measured since additional segmental material in closed syllables or words with other stress patterns (e.g., antepenultimate stress) may allow for later peak alignment; for the analysis of tonal peak height, all tokens were measured since the height of tonal peaks was not expected to be higher or lower in different tokens with different syllable types or stress patterns. Recordings were made with a *Sharp* MD-SR60 minidisc recorder and *Shure* 512 head-mounted microphone. The recordings were transferred to the computer at a 44.1 kHz sampling rate (16 bit) and analyzed in *Praat* (Boersma and Weenink, 1992–2011).

3.3. Measurements

For this data set, two types of measurements were taken: tonal peak alignment and tonal peak height. To do so, first each sentence was segmented into syllables in order to correctly identify the boundaries of the stressed syllable. Segmentation was carried out using the spectrogram, intonation (F0) and amplitude contours available in *Praat*, as well as by auditory inspection. To find the highest point during or after the stressed syllable in the intonation contour, the target word and following unstressed syllable(s) were highlighted, and the 'get maximum pitch' (F0) function in *Praat* was used.

Once the location of the peak was annotated, the alignment and height measurements could be made. First, the temporal position of both the tonal peak and the end of the stressed syllable were measured (in ms) in relation to the beginning of the utterance; the tonal peak alignment was calculated as the $\text{Time}_{\text{Peak}} - \text{Time}_{\text{SyllableEnd}}$. That is, positive values indicate a post-tonic peak, while negative values indicate that the peak was realized during the stressed syllable. A zero value, as appears in the subsequent figures, indicates the end of the stressed syllable. Also note that no peaks were observed prior to the beginning of the stressed syllable. Next, the tonal peak height was measured as the F0 height (in Hz) at the F0 maximum (as previously described above).

Since the effects of focus (on the subject) may appear not only on the subject itself, but also on the remainder of the utterance as well, measurements of tonal peaks related to the stressed syllable were taken in all three positions in the utterance (subject, verb and object). Statistical analysis was carried out per speaker for differences in alignment and height of peaks in the subject contrastive focus vs. broad focus conditions (e.g., alignment of the peak on the subject as $\text{Subject}_{\text{SCF}}$ vs. $\text{Subject}_{\text{BF}}$); given the small number of data points for a given position in the utterance (e.g., Subject) per speaker, the statistical analysis used was a Mann–Whitney *U* nonparametric test of mean ranks, where the z-statistic was used to

determine the significance level. Probability values reported at $p < 0.05$ are taken as statistically significant, although larger data sets with other statistical measures are needed to support the preliminary observations made here.

3.4. Hypotheses

The prediction is that Peruvian Spanish speakers will display prominence-lending features in the expression of focus, similar to other varieties of Spanish. In this study, only utterances with contrastive focus on the subject (labeled SCF) are examined. For these SCF utterances, the following features are predicted: (1) earlier alignment of the prenuclear peak on the subject under focus, and (2) a higher subject peak when this constituent receives focus and lower subsequent peaks on the verb and object. If these features are not observed, the speakers' productions of utterances in contrastive focus will be similar to those in broad focus. Also, Cuzco speakers with knowledge of Quechua (native Quechua speakers and Quechua-Spanish bilinguals) will show either no or fewer features used to express focus via intonation as a result of influence from Quechua, which marks focus in the morphology.

4. Results

Results presented in this section demonstrate that some intonation cues for focus can be observed for these Lima and Cuzco speakers, although variation between and within groups is found. The results are organized as follows. First, we view the intonation contour of several utterances in order to qualitatively observe the alignment and height of peaks and their relation to the stressed syllable. Next, the findings for tonal peak alignment are presented for Lima speakers and then Cuzco speakers, followed by the findings for tonal peak height. Finally, these results are combined in order to discuss patterns of behavior across groups.

4.1. A qualitative examination of tonal peaks in contrastive vs. broad focus

As can be seen in the example in Fig. 1 taken from a native Spanish speaker from Lima, the peak on the subject with contrastive focus in (b), as shown in small capitals in the figure gloss, is earlier and higher than the peak under broad focus in (a). Also, the post-focal peaks on the verb and object appear lower in the contrastive focus condition. In Fig. 2, an example from a Cuzco native Spanish speaker, the initial peak under contrastive focus also is higher and earlier. The subsequent peaks, however, do not appear to be markedly lower than in the broad focus condition. As previously indicated, in the following sections the utterances with contrastive focus will be analyzed according to tonal peak alignment and tonal peak height.

4.2. Peak alignment under contrastive focus

4.2.1. Lima speakers

The results show that the alignment of peaks in utterances with contrastive focus on the subject tends to be earlier than that of corresponding peaks in broad focus utterances. As previously mentioned, since the pragmatic condition may affect both the element directly under focus, the subject, and also the subsequent peaks, the alignment of the subject, verb, and object have been examined. Fig. 3(a–f) shows the findings for Lima speakers of peak alignment for utterances with broad

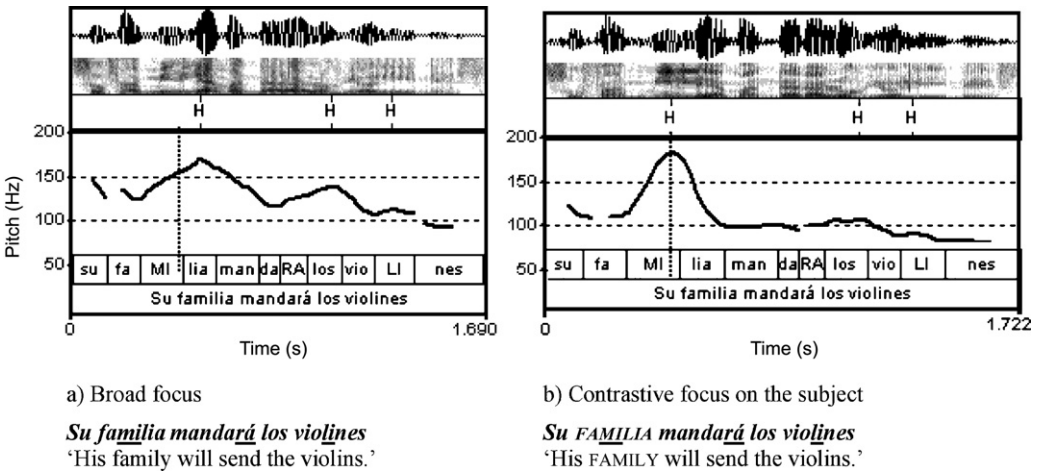
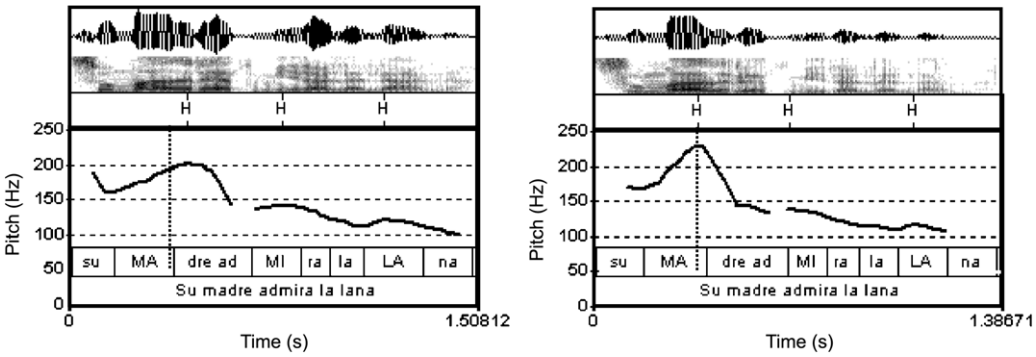


Fig. 1. A comparison between broad focus and contrastive focus contours. Contour (a) corresponds to a broad focus F0 contour; Contour (b) corresponds to an utterance with contrastive focus on the subject. Stressed syllables are underlined; subject under contrast shown in small capital letters. Lima native Spanish speaker (L01).

Notice the peaty on the v & o for Lima BF, but not in Cusco.



a) Broadfocus

Su madre admira la lana.
'Her mother admires the wool.'

b) Contrastive focus on the subject

Su MADRE admira la lana.
'Her MOTHER admires the wool.'

Fig. 2. A comparison between broad focus and contrastive focus contours. Contour (a) corresponds to a broad focus F0 contour; Contour (b) corresponds to an utterance with contrastive focus on the subject. Stressed syllables are underlined; subject under contrast shown in small capital letters. Cuzco native Spanish speaker (C08).

focus (BF) and contrastive focus on the subject (SCF). For each figure, the offset of the stressed syllable is shown as zero; the peaks occurring after the stress syllable offset show positive values and appear to the right of the dashed vertical line at zero, while peaks occurring during the stressed syllable show negative values and appear to the left of this vertical line. Statistical analysis was carried out at the individual level since each speaker's productions under contrastive focus were compared to those produced with broad focus. As previously noted, a Mann–Whitney *U* nonparametric test of mean ranks has been employed since the analysis is based on an unpaired number of data points and a normal distribution is not assumed for this small data set. A summary of the results of peak alignment for Lima speakers is found in Table 1; the results for Cuzco speakers are shown in Table 2 (see O'Rourke, 2005 for data tables). obj is sig, but prob not meaningful looking at the plot

Table 1
Comparison of peak alignment in contrastive focus (CF) and broad focus (BF) conditions for Lima speakers. Results from nonparametric Mann–Whitney *U* test for mean differences of mean rank between pragmatic conditions.

Alignment CF vs. BF				
Origin	Speaker	S	V	O
Lima	L01	*	*	nsd
	L02	*	*	nsd
	L03	*	nsd	nsd
	L04	*	*	*
	L05	nsd	nsd	nsd
	L06	*	*	*

* Significance at the $p < 0.05$ level or below.
nsd = no significant difference.

Table 2
Comparison of peak alignment in contrastive focus (CF) and broad focus (BF) conditions for Cuzco speakers. Results from nonparametric Mann–Whitney *U* test for mean differences of mean rank between pragmatic conditions.

Alignment CF vs. BF				
Origin	Speaker	S	V	O
Cuzco	C01	(*)	*	nsd
	C08	*	nsd	*
	C09	*	*	*
	C21	(*)	nsd	nsd
	C23	nsd	nsd	*
	C33	nsd	*	nsd

* Significance at the $p < 0.05$ level or below.
nsd = no significant difference. The remainder of the Cuzco speakers showed no significant differences in peak alignment in any of the positions.

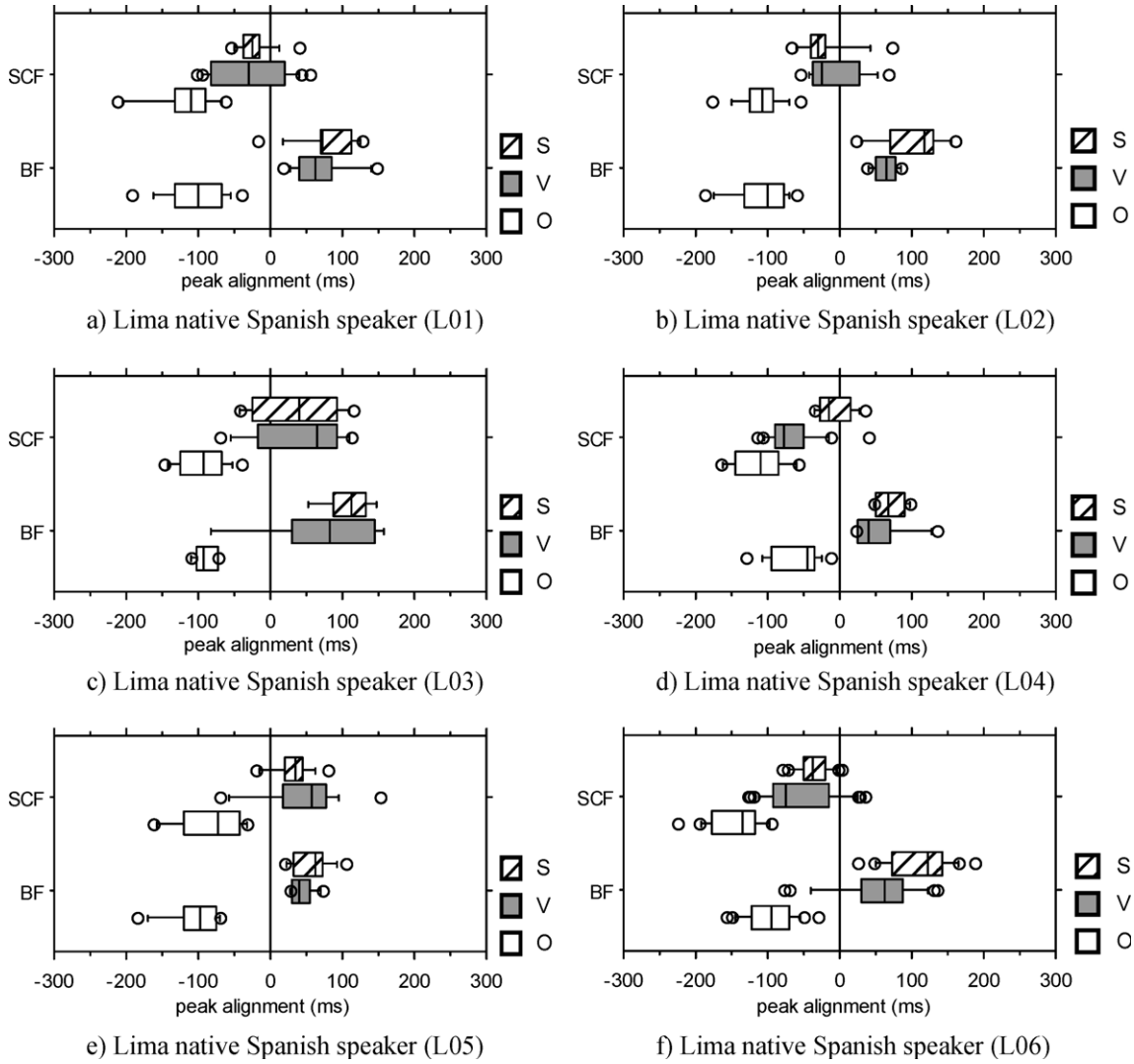


Fig. 3. Alignment of peaks (in ms) in open syllables in words with penultimate stress. Vertical dashed line at zero represents stressed syllable offset. S = subject in initial position (shown in striped diagonal lines), V = verb in medial position (shown in grey), O = object in utterance final position (shown in white). Box plots represent a minimum of 5 data points. BF = utterances with broad focus; SCF = utterances with contrastive focus on the subject in initial position.

As shown in Fig. 3, the peak associated with the stressed syllable in the subject is aligned earlier under contrastive focus. For five of these speakers (L01, L02, L04, and L06), the peak shifts in position, on average, from the post-tonic in broad focus to aligning within the stressed syllable in contrastive focus. In subject position, this change is shown to be significant ($p < 0.05$ or less). Speakers L03 and L05 show post-tonic peaks for both contrastive focus and broad focus conditions; while the peaks in contrastive focus on the subject are earlier for both speakers, these differences reach significance for speaker L03 ($p < 0.05$) but not for speaker L05 ($p = 0.0724$). In addition, for some speakers the peaks associated with the verb and object are earlier in SCF utterances (see Table 1 for summary). Overall, these results correspond to findings previously reported for other varieties, including Madrid Spanish (Face, 2001), Peninsular Spanish (de la Mota Gorriz, 1997), and Porteño Spanish (Barjam, 2004).

A secondary alignment effect is also observed: the prenuclear peak associated with the verb likewise aligns within the tonic syllable, even though the verb itself is not considered to be under contrastive focus. This behavior is observed with speakers L01, L04, and L06 ($p < 0.0001$) and L02 ($p = 0.0016$), but is not shown for the remaining two speakers. Likewise, the nuclear peak for two speakers is aligned significantly earlier in the utterances with contrastive focus on the subject (L04, $p = 0.0039$; L06, $p = 0.0002$). For those speakers who show earlier focal and postfocal alignment of peaks within the stressed syllable, there appears to be a type of inward movement of peak alignments; that is, the peak on the initial subject is farthest to the right (i.e., latest with relationship to the stressed syllable), followed by the peak on the medial verb, which is realized earlier, followed by the nuclear peak on the object, which is the earliest of the three. Examination of longer utterances may

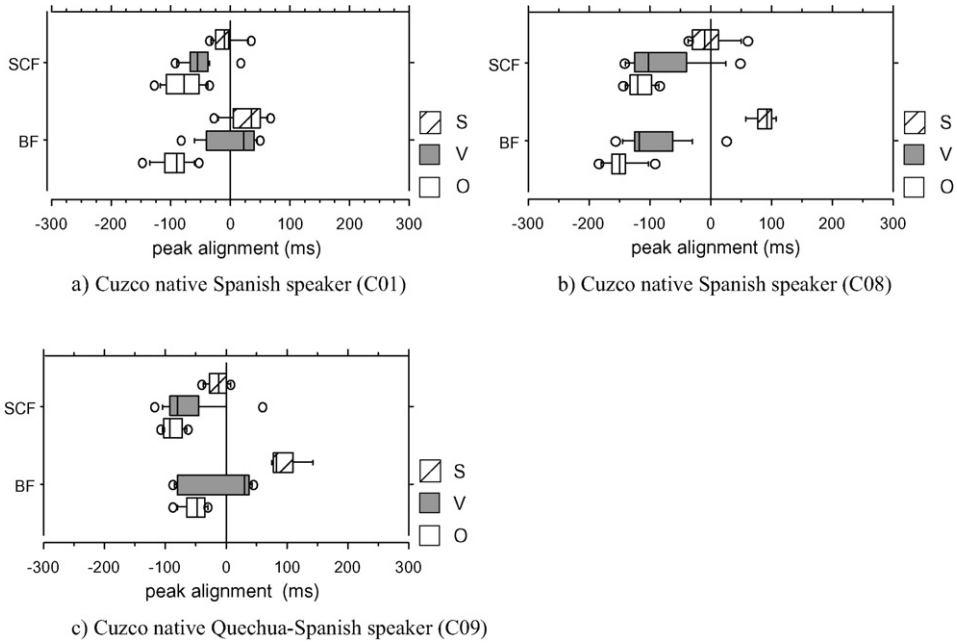


Fig. 4. Alignment of peaks (in ms) in open syllables in words with penultimate stress. Vertical dashed line represents stressed syllable offset. S = subject in initial position (shown in striped diagonal lines), V = verb in medial position (shown in grey), O = object in utterance final position (shown in white). Box plots represent a minimum of 5 data points. BF = utterances with broad focus; SCF = utterances with contrastive focus on the subject in initial position.

further demonstrate the extent of this phenomenon (e.g., if it occurs within complex subject phrases and verb phrases). The degree to which this behavior is simply phonetic, or the extent to which it may be phonological, would also need to be determined.[10] However, further exploration of this issue goes beyond the scope of this discussion.

4.2.2. Cuzco speakers

For Cuzco speakers, the findings on peak alignment of the subject in contrastive focus may be divided first into two broad categories, those speakers that show differences in alignment (6 speakers) and those that do not (11 speakers). Initially recordings from fifteen Cuzco Spanish speakers were examined: seven native Spanish speakers, five Quechua-Spanish bilinguals, and three native Quechua speakers. However, only native Spanish speaker C01 showed potential alignment differences on the subject and on either the verb and/or object, similar to the patterns reported for Lima (see Fig. 4(a)). In order to corroborate these findings, the recordings of two additional male speakers from Cuzco were examined (see section 3.1 for a description of all study participants). Preliminary auditory examination of their recordings indicated that these speakers did place prominence on the subject under contrastive focus.[11] These participants were native Spanish speakers from Cuzco (labeled C08 and C09). For both speakers, utterances in broad focus showed an average post-tonic peak alignment only for the subject; the mean prenuclear peak alignment on the verb was within the stressed syllable (see Fig. 4(b and c)).

The results show that in utterances with contrastive focus on the subject, peaks for both the subject and the verb align within the stressed syllable. In subject position, this difference is significant for two speakers (C08, $p = 0.0074$; C09, $p = 0.0062$), and approaches significance for one speaker (C01, $p = 0.0506$). Also, similar to several Lima speakers, peaks associated with the verb are aligned significantly earlier for two Cuzco speakers (C01, $p = 0.0025$; C09, $p = 0.0180$). The results for the object show more variation: for one speaker, the peak is significantly earlier (C09, $p = 0.0016$), for another it is significantly later in the stressed syllable (C08, $p = 0.0128$), and for another, there is no significant difference (C01, $p = 0.2270$). The findings for these speakers are summarized in Table 2.

[10] An initial proposal was that this behavior could be a general reduction of articulatory effort used along the length of the utterance, resulting in subsequent peaks that align earlier than previous peaks. However, as pointed out by one reviewer, the time-scale over the utterance is quite large in comparison to the much smaller differences in peak alignment under BF and CF conditions. Rather, another possible explanation may be related to distance between peaks in that postfocal peaks can align earlier since there is more room after the tonic-aligned peak under focus (i.e., a lack of time-compression between peaks). One problem with this explanation is that it would predict a sensitivity to the number of intervening unstressed syllables, which is not necessarily expected and would need to be further tested.

[11] These speakers have been analyzed as illustrative examples of speakers, like speaker C01, who use prosodic features in the expression of focus. In this way, the behavior of speaker C01 is corroborated. While the inclusion of these speakers helps to explore this issue further, ideally recordings of more Quechua-Spanish bilinguals and native Quechua speakers are needed to balance the number of observations per group.

wow! very clear plot

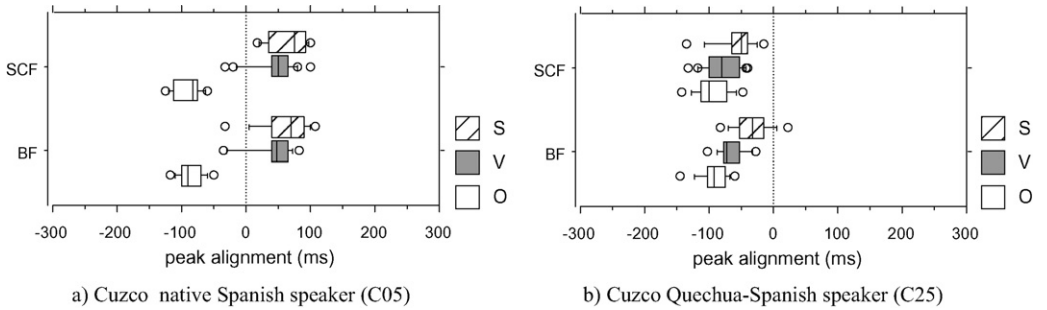


Fig. 5. Representative graphs of speakers with no significant difference in peak alignment between utterances with contrastive focus on the subject (SCF) and utterances with broad focus (BF). Alignment of peaks (in ms) in words with open syllables and penultimate stress. Vertical dashed line represents stressed syllable offset. S = subject in initial position (shown in striped diagonal lines), V = verb in medial position (shown in grey), O = object in utterance final position (shown in white). Box plots represent a minimum of 5 data points. BF = utterances with broad focus; SCF = utterances with contrastive focus on the subject in initial position.

this is good place to talk about significance testing – we can take "approaching" sig to be in support of us or against us

In addition to the speakers described above who show significant differences (or nearly so) in more than one position within the utterances, there are three speakers who show just one alignment distinction: for one bilingual speaker, the difference approaches significance on the subject (C21, $p = 0.0543$); one bilingual shows significance on the object (C23, $p = 0.0367$); and one Quechua native speaker shows a significant difference on the verb (C33, $p = 0.0269$) (see Table 2 for summary).

The remaining speakers did not show any significant differences in peak alignment between pragmatic conditions.[12] See Fig. 5 for examples of a lack of alignment differences. Fig. 5(a) shows post-tonic prenuclear peaks in both pragmatic conditions; Fig. 5(b) shows tonic prenuclear peaks in both conditions.

4.3. *Peak height in utterances with contrastive focus on the subject*

In the previous sections, the alignment of tonal peaks with respect to the offset of the stressed syllable was examined. In this section, the height of tonal peaks will be addressed, followed by the relationship between peaks in a given utterance. As previously noted, in this discussion, words with all three stress patterns are included (antepenultimate, penultimate and final stress), as well as words with both open and closed stressed syllables since these factors are not expected to affect the ultimate height attained in these utterances.

4.3.1. *Lima speakers*

The results for Lima speakers can be seen in Fig. 6, which shows the peak height observed in utterances with subject contrastive focus and broad focus. The graphs in Fig. 6(a–f) show that the initial peak on the subject is sometimes significantly higher in contrastive focus (L01, $p = 0.0089$; L06, $p < 0.0001$), whereas in other cases, it is not. More often, either the verb and/or the object is significantly lower than in broad focus: significantly lower verbal peaks appear with speakers L01, L02, L04, L06; significantly lower object peaks are found with speakers L01, L02, L04, L05, L06. Speaker L03 does not show significant differences in height between the two focus conditions. See Table 3 for a summary of these findings.

4.3.2. *Cuzco speakers*

The data for Cuzco speakers can be broadly divided according to those who demonstrated some difference in peak height (11 speakers), and those who did not show any differences (6 speakers). Of these eleven Cuzco speakers, one person (speaker

Table 3
Comparison of peak height in contrastive focus (CF) and broad focus (BF) conditions for Lima speakers. Results from nonparametric Mann–Whitney *U* test for mean differences of mean rank between pragmatic conditions.

Height CF vs. BF				
Origin	Speaker	S	V	O
Lima	L01	*	*	*
	L02	nsd	*	*
	L03	nsd	nsd	nsd
	L04	nsd	*	*
	L05	nsd	nsd	*
	L06	*	*	*

* Significance at the $p < 0.05$ level or below.
nsd = no significant difference. Note that peaks were significantly higher in subject position under subject contrastive focus, but significantly lower on the verb and object in utterances with subject contrastive focus.

so subject peak doesn't change much – all alignment, but post-focal content gets deaccented/lowe peak

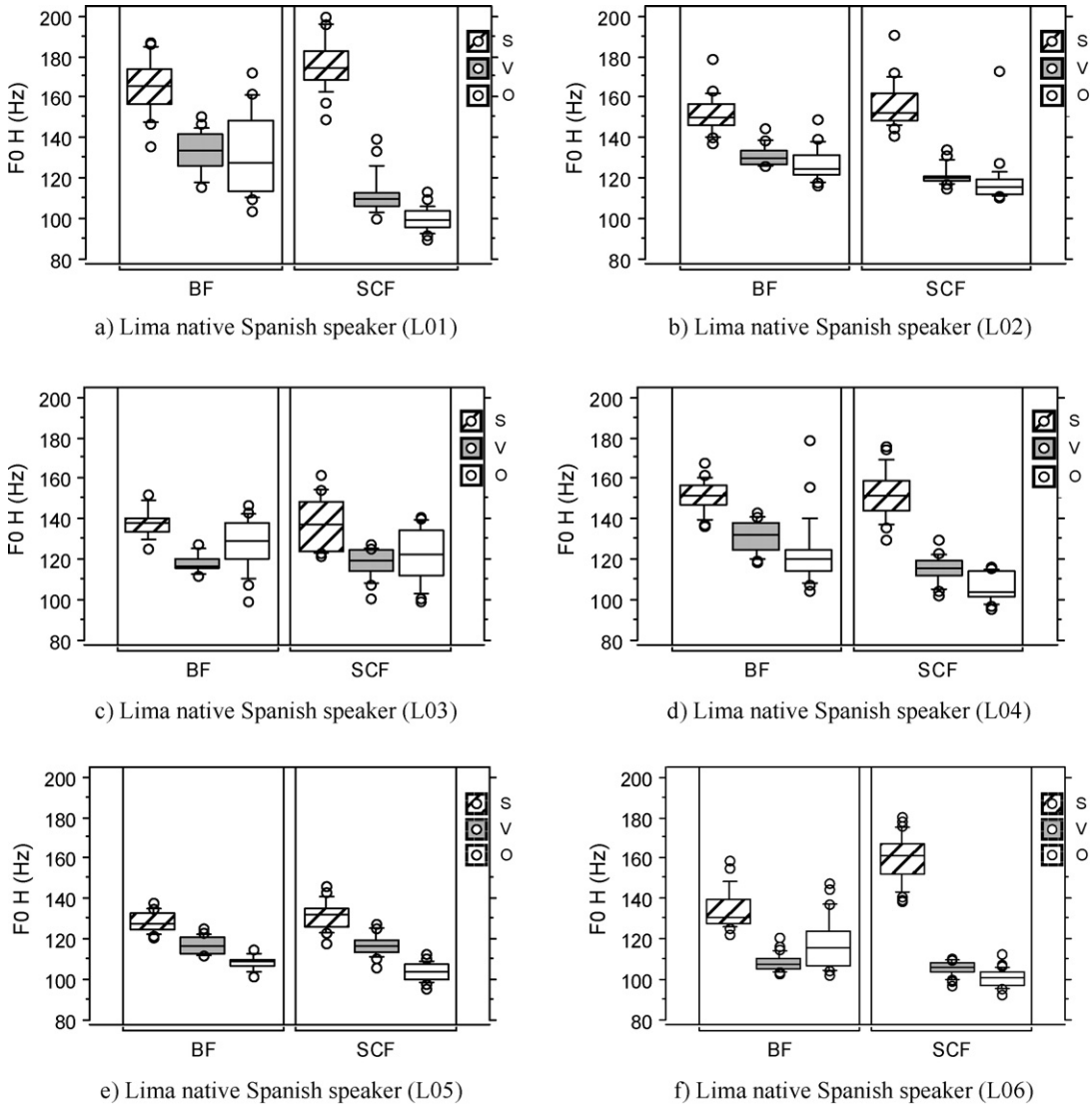


Fig. 6. Height of peaks (in Hz). Speakers L01 and L06 show significantly higher peaks for subjects with contrastive focus. Also, speakers L01, L02, L04, L06 show significantly lower peaks on the verb and object in utterances with contrastive focus on the subject. Speaker L05 shows the same behavior only on the object in utterance-final position. No significant differences are found for speaker L03. S = subject in initial position (shown in striped diagonal lines), V = verb in medial position (shown in grey), O = object in utterance final position (shown in white). Box plots represent a minimum of 5 data points. BF = utterances with broad focus; SCF = utterances with contrastive focus on the subject in initial position.

C09) shows a significantly higher peak height on the subject, similar to Lima speakers L01 and L06. This Cuzco speaker also shows significantly lower average peak heights for the verb and object positions. This speaker also previously demonstrated a difference in subject peak alignment between pragmatic conditions (see Fig. 4(c)). Similarly, there are three speakers who show significantly lower peaks on the verb and/or object in contrastive focus, while maintaining the same average height on the subject in both conditions: significantly lower verbal peaks appear with speakers C01, C06, and C21; significantly lower object peaks are shown with speakers C06 and C21) (see Fig. 7 and Table 4(a) for summary).

There is another group of speakers who demonstrate significantly lower peak heights on the subject in the contrastive focus condition. This behavior is contrary to expectations, since a higher peak is the expected prominence-lending feature used to enhance the subject under focus. If lower peaks are also somehow more 'prominent' by being markedly lower than

[12] Another potential effect of contrastive focus is an increased duration of the stressed syllable under focus, which would in turn affect the extent to which a peak is considered 'early' or 'late' (see Face, 2001 for discussion). However, in O'Rourke (2005), no significant durational differences were found between contrastive focus on the subject and broad focus for Lima speakers (with the exception of one speaker, L04); also, no significant differences were found for Cuzco speakers. This finding suggests that duration is not used as a primary signal for focus in these varieties.

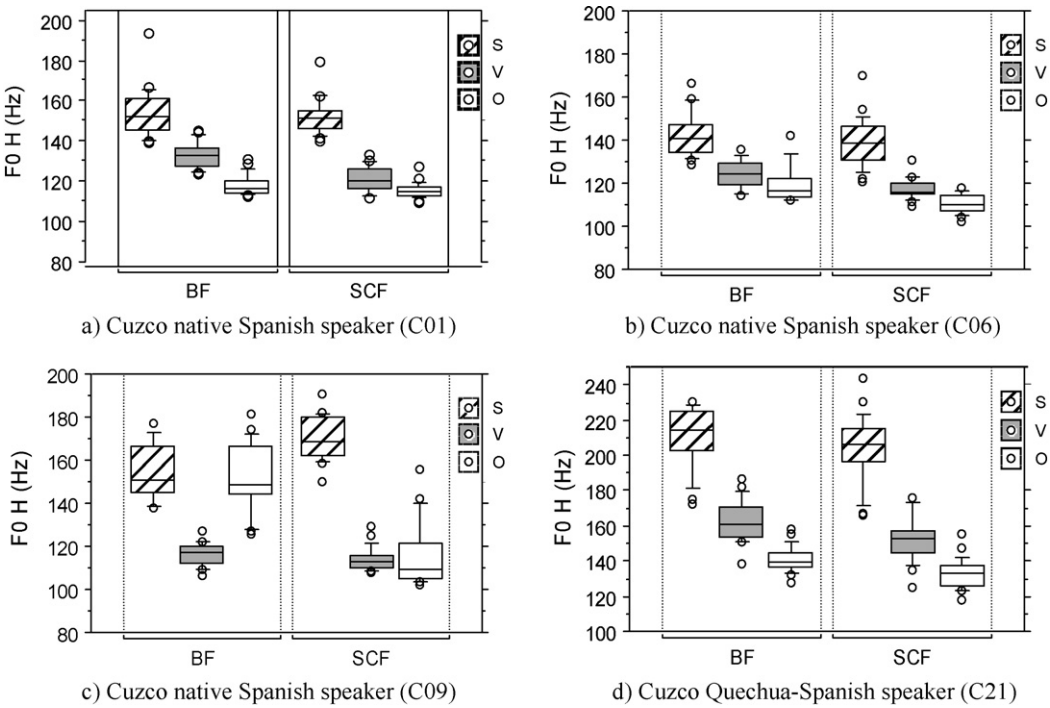


Fig. 7. Height of peaks (in Hz). Speaker C09 shows a significantly higher average peak height on the subject in contrastive focus and significantly lower peaks on the verb and object. Significantly lower peaks appear on the verb (speakers C01, C06, and C21) and on the object (speakers C06 and C21) in utterances with contrastive focus, while the peak on the subject is not significantly different for these speakers. S = subject in initial position (shown in striped diagonal lines), V = verb in medial position (shown in grey), O = object in utterance final position (shown in white). Box plots represent a minimum of 5 data points. BF = utterances with broad focus; SCF = utterances with contrastive focus on the subject in initial position. Corresponding peak height pattern for broad focus declaratives is listed in parentheses for reference.

usual, this type of strategy will need to be further tested to examine if Cuzco speakers actually perceive these lower peaks as under focus as well.

Fig. 8 and Table 4(b) show the results for the speakers in this group. Many of these same speakers also produced significantly lower peaks in the verb and object positions, suggesting that they were 'sensitive' to the test conditions in that they produced different types of response contours according to the context presented (broad focus vs. contrastive focus on the subject). Compared to those speakers listed in Table 4, the remaining six speakers (C02, C03, C04, C08, C24, and C33) show no significant difference in peak height in the responses produced to the broad focus and contrastive focus elicitation questions (see examples in Fig. 9).

Table 4
Comparison of peak height in contrastive focus (CF) and broad focus (BF) conditions for Cuzco speakers. Results from nonparametric Mann–Whitney *U* test for mean differences of mean rank between pragmatic conditions.

Height CF vs. BF				
Origin	Speaker	S	V	O
(a) Significantly higher peaks on subject; and/or significantly lower peaks on verb and object				
Cuzco	C01	nsd	*	nsd
	C06	nsd	*	*
	C09	*	nsd	*
	C21	nsd	*	*
(b) Significantly lower peaks on subject; in some cases, also significantly lower peaks on verb and object				
Cuzco	C05	*	nsd	*
	C07	*	*	*
	C22	*	nsd	*
	C23	*	*	nsd
	C25	*	*	*
	C31	*	*	nsd
	C32	*	*	*

* Significance at the $p < 0.05$ level or below.
nsd = no significant difference. The remainder of the Cuzco speakers showed no differences in peak height between pragmatic conditions.

21-25 = bilingual

31-33 = Quechua

notice that for CF, even if there's no sig diff, the F0 is lower than BF

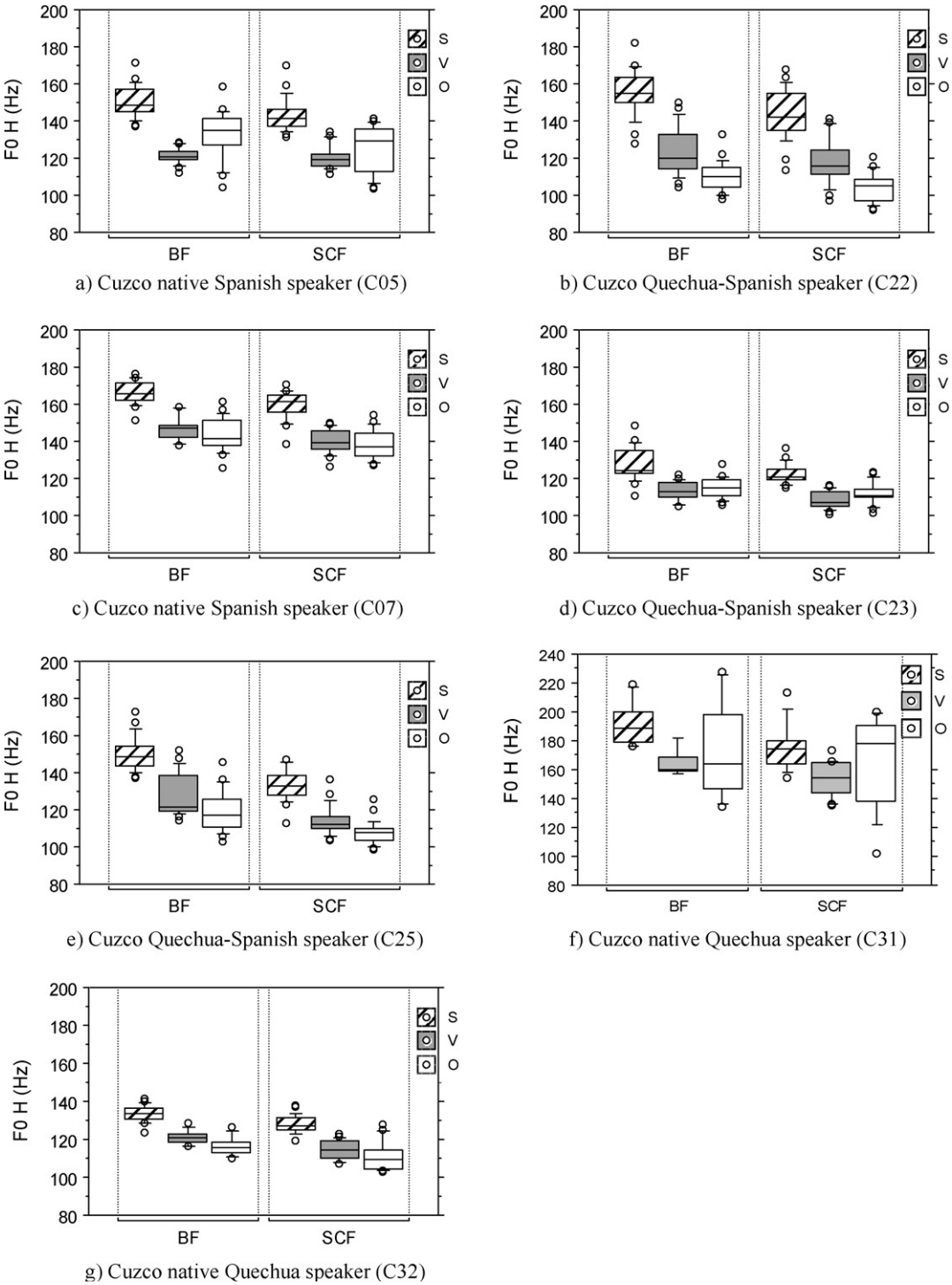


Fig. 8. Cuzco speakers with significantly lower average peak height on the subject in contrastive focus. Height of peaks (in Hz). S = subject in initial position (shown in striped diagonal lines), V = verb in medial position (shown in grey), O = object in utterance final position (shown in white). Box plots represent a minimum of 5 data points. BF = utterances with broad focus; SCF = utterances with contrastive focus on the subject in initial position.

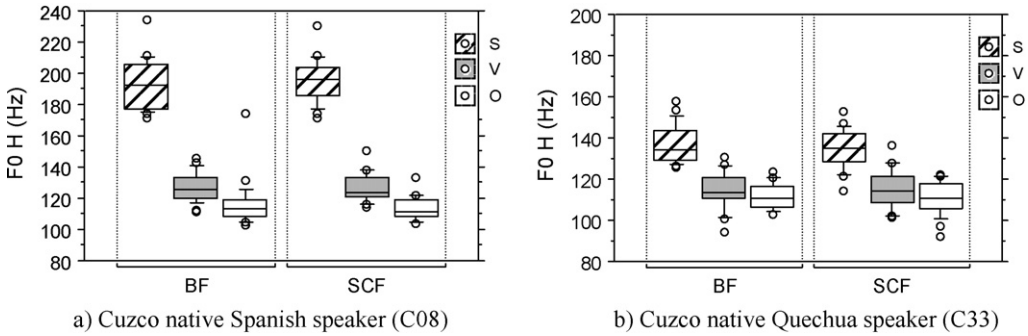


Fig. 9. Representative graphs of speakers with no significant differences in peak height between utterances with contrastive focus on the subject (SCF) and utterances with broad focus (BF). Height of peaks (in Hz). S = subject in initial position (shown in striped diagonal lines), V = verb in medial position (shown in grey), O = object in utterance final position (shown in white). Box plots represent a minimum of 5 data points. BF = utterances with broad focus; SCF = utterances with contrastive focus on the subject in initial position. Corresponding peak height pattern for broad focus declaratives is listed in parentheses for reference.

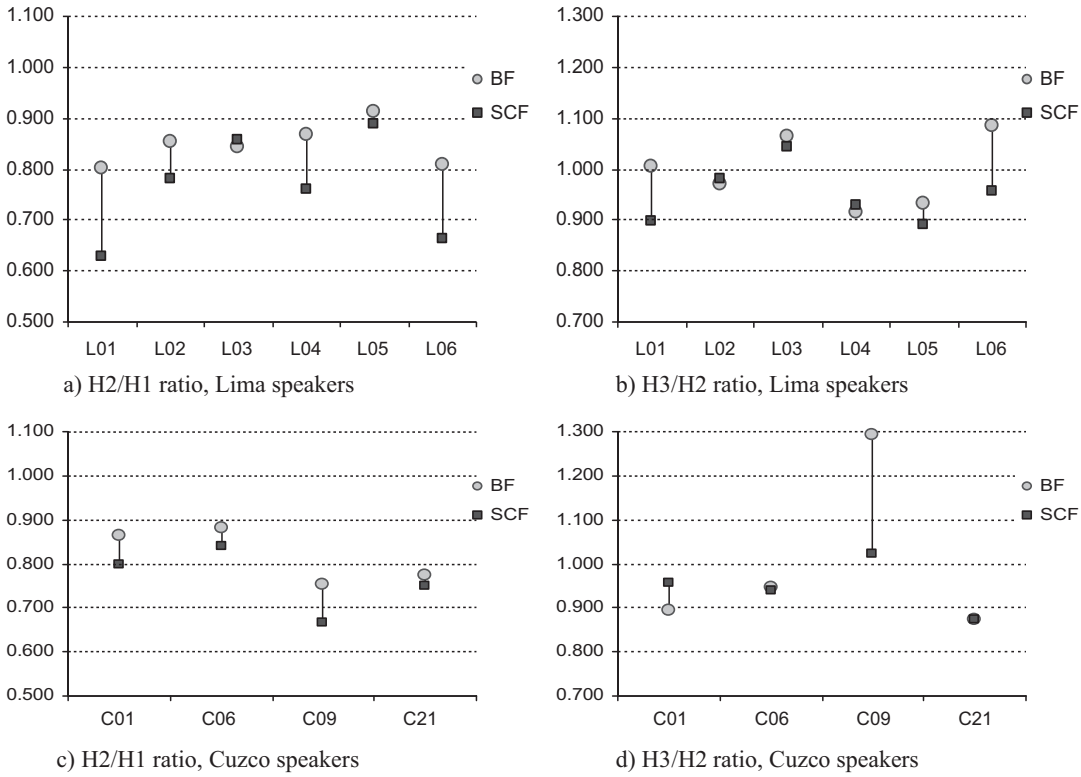


Fig. 10. Ratio of peaks for Lima and Cuzco speakers in broad focus (BF) and subject contrastive focus (SCF) utterances. Values less than one indicate a lower subsequent peak; values greater than one indicate a higher subsequent peak.

display this graph & critique, but we don't need to actually go over the data here

4.3.3. Peak height relationships in two focus conditions

To examine further the relationship between peak height in broad focus and subject contrastive focus, a ratio of adjacent peak heights was calculated for both utterance types. This calculation indicates the height of a peak relative to the height of the previous peak, i.e. a localized downstep ratio. A similar calculation is found in Sityaev and House (2003) for the scaling of peaks in Southern British English.[13] For this data set, the height of the second peak is divided by the height of the first peak, H2/H1, and similarly, H3/H2. The resulting values that are less than one indicate a lower second peak. The lower the ratio, the lower the second peak is compared to the first, i.e. the greater the downstep.

The graphs in Fig. 10 show the ratio of peaks for Lima speakers in (a) and (b), and for Cuzco speakers in (c) and (d). Only those Cuzco speakers who demonstrated significant differences in peak height in the previous section were included in this

[13] See also Prieto (1998) for an examination of downstep and tonal scaling in Spanish which includes the peak and preceding low tone.

calculation (those appearing in Fig. 7). Those speakers that show nearly the same ratio for adjacent peaks (e.g., speaker L03, H2/H1 ratio), demonstrate a similar degree of downstep between the two peaks in both pragmatic conditions. Those speakers who show a lower ratio for contrastive focus on the subject (SCF), display a greater degree of downstep to the subsequent peak, in comparison to broad focus (BF) (e.g., speaker L01, H2/H1 ratio). A Mann–Whittney *U* test of mean ranks of peak ratios was conducted between pragmatic conditions. Lima speakers L01, L02, L04 and L06 show significant differences for H2/H1 (Fig. 10(a)); for H3/H2, Lima speakers L01 and L06 show significant differences (Fig. 10(b)); Cuzco speakers C01, C06 and C09 show significant differences for H2/H1 (Fig. 10(c)); and for H3/H2, Cuzco speakers C01 and C09 show significant differences in peak height ratios (Fig. 7(d)). The remaining Cuzco speakers show no significant differences across conditions.

4.4. Comparison of focus strategies

Given the same task, Lima speakers tended to produce prosodic distinctions more consistently as a group between broad focus and contrastive focus conditions than Cuzco speakers. For example, five of the six Lima speakers showed earlier alignment of the tonal peak under contrastive focus, and four of these also showed earlier tonal alignment of post-focal peaks in the utterances with contrastive focus (see Table 1). In comparison, of the seventeen Cuzco speakers, only two speakers showed similar early tonal alignment on the subject under focus, with two more approaching a significant difference. Post-focal earlier peak alignment for contrastive focus utterances was observed with these two speakers, as well as with three other Cuzco speakers (see Table 2). In terms of tonal peak height, four of the six Lima speakers showed lower post-focal peaks, and two of these also showed a higher peak on the subject under focus (see Table 3). Of the Cuzco speakers, thirteen showed lower post-focal peaks; of these only one showed a higher peak on the subject under contrastive focus, while three showed no difference in tonal peak height between pragmatic conditions, and seven showed lower peaks on the subject under contrastive focus (see Table 4). This last behavior, as mentioned, while not initially expected, may demonstrate another way in which the entire utterance is signaled as being a marked condition.

To summarize the information given in this paper, in terms of strategies for marking contrastive focus, the Lima speakers appear to employ earlier peak alignment, often in conjunction with lower post-focal peaks and sometimes higher tonal peaks under focus. Cuzco speakers seem to employ lower post-focal peaks, while most frequently a lower tonal peak under focus is used, followed by no distinction, and then a higher peak under focus. While the above description characterizes the behavior observed, the next step would be to determine how essential these features (tonal peak alignment and height) are in signaling and perceiving focus. The relative importance of each feature used alone or in combination may differ across varieties. This type of analysis, which would give values and weights to each of the features, in order to provide a type of 'focus index', lies beyond the scope and design of the present study.

An examination of speaker behavior with respect to origin and knowledge of Quechua shows a greater degree of variability among Cuzco speakers compared to Lima speakers. Within Cuzco we see that some native Spanish speakers (C01, C08, C09) show behavior similar to Lima speakers with respect to peak alignment, while the remaining six Cuzco native Spanish speakers do not. Of the Cuzco speakers with knowledge of Quechua, the majority tend not to show alignment distinctions in any of the positions (with the exception of the bilinguals C23, and C21 to some extent, and native Quechua speaker C33 who show earlier alignment in one position). For the feature of peak height, there were also some Cuzco native Spanish speakers (C01, C06, and C09), who behaved similarly to Lima Spanish speakers, while two showed lower peaks on the subject (C05 and C07), and four others showed no tonal height distinctions between conditions. The bilinguals showed a similar degree of variation, with one speaker (C21) demonstrating patterns similar to Lima, three showing lower peaks on the subject (C22, C23, and C25), and one showing no height distinctions (C24). Finally, native Quechua speakers only show either lower peaks on the subject (C31 and C32) or no distinctions in height between contrastive and broad focus (C33).

Overall, there appears to be a trend towards non-Lima like behavior as we move from the native Spanish group, to Quechua-Spanish bilinguals, to native Quechua speakers. In order to gain a more in-depth view of these groups, the employment of one or more features could be described as a type of focus pattern. In addition, the speakers' knowledge of Quechua could be evaluated according to certain factors (e.g., their own fluency, their parents' language, their language use in the community, and their level of education), and then assigned a rating or Quechua 'language use index'.[14] The frequency or appearance of a given pattern within a particular group may then help to characterize one group's reliance on (or preference for) a certain pattern in a given context (e.g., formal vs. informal).[15] However, like the 'focus index' described above, this 'language use index' would be most illustrative if the relative importance of each language factor were determined since each factor may not contribute equal weight in terms of the overall speaker behavior.

5. Conclusions

This paper has presented the results of an experiment on contrastive focus in Spanish as spoken in two regions of Peru, Lima and Cuzco. The participants in the Lima group were all native speakers of Spanish, whereas the Cuzco speakers were

[14] As a preliminary examination using this type of approach, see O'Rourke (2005) in which each language factor is assigned equal weight.

[15] See, for example, Escobar (1990) for an analysis of speaker behavior according to different systems of pronoun usage divided according to sociolinguistic variables (e.g., origin, education, social class, etc.).

either native Spanish speakers, Quechua-Spanish bilinguals, or native Quechua speakers. Variation in the number and type of focal strategies used in the expression of utterances with contrastive focus on the subject differed between Lima and Cuzco, and also within each of the groups, although Lima behaved relatively more uniformly. While most Lima speakers exhibited several prominence-lending features as predicted, only two Cuzco speakers followed these patterns while the remainder exhibited less focus features, no features, or features that were in the opposite direction of the prediction.

In terms of methodology, when given the same task as Lima speakers, Cuzco speakers demonstrated more reliance on tonal peak height distinctions. It should be noted that no additional highlighting of the focused word was shown (such as underlining, capitalization or bold). Further enhancement of the target utterance text may increase the possibility of more Cuzco speakers producing prominence-lending cues. A different type of task may also provide independent support for the findings presented here. To describe the variation observed, two types of indexes were discussed that could potentially be useful metrics to further characterize the patterns seen, although issues of weighting each contributing factor were also noted. In particular, the 'language use index' would help to understand speaker behavior, since intonation features do not necessarily pattern directly with the self-reported native language/s of the Cuzco speakers.

To summarize, Lima speakers in general and some Cuzco speakers employ features that lend prominence to the peak of the element under focus and de-emphasize the elements that are not focused, which is consistent with previous descriptions of Madrid and Peninsular Spanish. Higher focused peaks were more variable across speakers and groups, as was the case with previous reports on Spanish, although post-focal deaccenting was relatively more common, similar to Madrid Spanish. The remainder of Cuzco speakers showed less similarity with both Lima speakers and other descriptions of Spanish contrastive intonation. The tentative conclusion, then, is that Cuzco Spanish speakers in general do not express contrastive focus intonation in the same way as found in Lima and other Spanish varieties. These Cuzco speakers, who report possessing varying levels of bilingualism and contact with Quechua, may actually have a larger variety of linguistic resources available to them to express focus, which may account for the difference seen in the behavior between the speakers from these two groups. The findings reported here show the use of similar features being employed with differences in pragmatic meaning according to speaker (e.g., prenuclear tonic peak alignment is used in broad focus for some Cuzco speakers and in contrastive focus for other Lima and Cuzco speakers).[16] This study on contrastive focus helps identify how Spanish intonation may differ across dialects, including those in contact with other languages and intonation systems.

Acknowledgements

Funding for this research was received from the University of Illinois at Urbana-Champaign 'Summer Field Research Grant' from the Center for Latin American and Caribbean Studies and the 'Dissertation Travel Grant' from the Graduate College. I would like to thank José Ignacio Hualde and Anna María Escobar for their advice during the development of this project as well as Virginia Zavala Cisneros and Jean-Jacques Decoster in Lima and Cuzco respectively for their support in conducting these recordings. Last, I am grateful for the many insightful comments from two anonymous reviewers which helped to clarify certain explanations and concepts, and to improve the overall organization of the paper. Any error in analysis or design, however, remains my own.

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[16] Note that in Basque-influenced Spanish, early alignment was observed for broad focus (Elordieta, 2003); see Colantoni (2011) for discussion of features common in contact Spanish.

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